

Anaphylaxis



Section I: Scenario Demographics

Scenario Title:	Anaphylaxis with Laryngospasm
Date of Development:	(23/02/2016)
Target Learning Group:	☐ Juniors (PGY 1 – 2) ☒ Seniors (PGY ≥ 3) ☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Donika Orlich
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To expose learners severe anaphylaxis and laryngospasm complication during intubation.
CRM Objectives:	1) Displays leadership by maintaining calm demeanor during crisis and acting decisively 2) Employs good communication skills by using closed loop, listening to the input of others, and addressing concerned family members
Medical Objectives:	1) Implements the ED treatment of anaphylaxis 2) Predicts and prepares for difficult intubation 3) Demonstrates an approach to laryngospasm during intubation

Case Summary: Brief Summary of Case Progression and Major Events

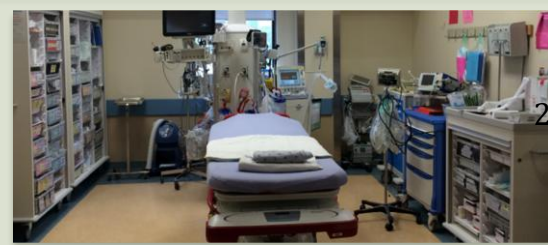
A 7-year-old male presents with wheeze, rash and increased WOB after eating a birthday cake. He has a known allergy to peanuts. The team must initiate usual anaphylaxis treatment including salbutamol for bronchospasm. The patient will then develop worsened hypotension, requiring the start of an epinephrine infusion. After this the patient will experience increased angioedema, prompting the team to consider intubation. If no paralytic is used for intubation (or if intubation is delayed), the patient will experience laryngospasm. The team will be unable to bag-mask ventilate the patient until they ask for either deeper sedation or a paralytic. If a paralytic is used, the team will be able to successfully intubate the child.

References

- Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.
- Hobaika AB, Lorentz MN. [Laryngospasm]. Rev Bras Anesthesiol. 2009 Jul-Aug;59(4):487-95. Review. Portuguese. PubMed PMID: 19669024
- Larson CP Jr. Laryngospasm—the best treatment. Anesthesiology. 1998 Nov;89(5):1293-4. PubMed PMID: 9822036
- <http://lifeinthefastlane.com/ccclaryngospasm/>
- Al-Metwalli RR, Mowafi HA, Ismail SA: Gentle chest compression relieves extubation laryngospasm in children. J Anesth 2010; 24:854–7. PMID: 20976504
- <http://first10em.com/2015/03/04/laryngospasm/>



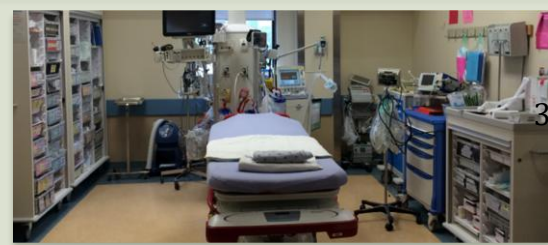
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Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Pediatric Computerized Mannequin	Realism: <i>Select most important dimension(s)</i>	☒ Conceptual
	☐ Mannequin		☒ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
Confederates	Brief Description of Role		
Mother	To help with airway history		
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☒ Needle cric supplies	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☐ Other:	
☒ Intraosseous Set-up	☐ LMA	☒ Other: Peds NRB PEEP valve	
D. Moulage			
Rash over arms and legs.			
E. Approximate Timing			
Set-Up:	3 min	Scenario:	15 min
		Debriefing:	30 min

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Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

A 7-year-old boy arrives via EMS with increased work of breathing. He has a known allergy to peanuts and developed symptoms after eating birthday cake at a party. He has been given 0.15mg IM epinephrine 10 minutes ago by his mother. Current vital are: HR 140, BP 85/60, RR 40, O2 98% on NRB. He has some ongoing wheeze noted by EMS.

B. Patient Profile and History

Patient Name: Cayden McFadden	Age: 7	Weight: 28 kg
Gender: ☒ M ☐ F	Code Status: Full.	

Chief Complaint: Allergic Reaction

History of Presenting Illness: Ate cake at a party and developed wheeze, shortness of breath and rash within 10 minutes. Mother administered his epinephrine auto-injector and called EMS.

Past Medical History:	Allergy to peanuts	Medications:	Epinephrine auto-injector

Allergies: Peanuts.

Social History: Lives with parents.

Family History: Non-contributory.

Review of Systems:	CNS:	Nil		
	HEENT:	No facial/mouth swelling.		
	CVS:	Nil.		
	RESP:	Audible wheeze.		
	GI:	Complaining of stomach pains. No vomiting.		
	GU:	Nil.		
	MSK:	Mil	INT:	Rash over entire body.

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display	☒ Monitor On, no data displayed	☐ Monitor on Standard Display
HR: 130/min	BP: 80/50	RR: 40/min
O ₂ SAT: 97% NRB		
Rhythm: Sinus	T: 36.1°C	Glucose: 6.2 mmol/L
GCS: 15 (E4 V5 M6)		
General Status: Unwell		
CNS:	Alert, oriented. Appears scared.	
HEENT:	No facial/airway edema.	
CVS:	Normal HS. No murmur.	
RESP:	Audible wheeze. Chest tight. Tracheal tug. Intercostal indrawing.	
ABDO:	Soft, diffuse discomfort. No rebound. No peritonitis	
GU:	Nil.	
MSK:	Nil	SKIN: Rash over entire body.

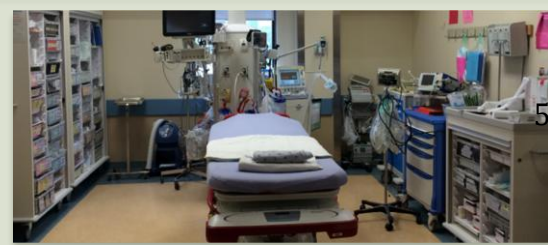
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Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: NSR HR: 120/min BP: 80/50 RR: 40/min O ₂ SAT: 97% NRB T: 36.1°C	Significant work of breathing. Mother panicked at bedside.	<u>Learner Actions</u> - ☐ Monitors, O ₂ , IV access - ☐ IM epinephrine left lateral thigh at 0.01mg/kg - ☐ IV Benadryl (1mg/kg) - ☐ IV Ranitidine (1mg/kg) - ☐ IV Methylprednisolone (1-2mg/kg) - ☐ IV NS bolus (20mL/kg) - ☐ Salbutamol (5mg) and Ipratropium (0.5mg) in 3mL NS neb - ☐ Cap glucose - ☐ Portable CXR	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - Epi → HR 130, BP 85/50 - NS bolus → HR 125, BP 85/55 - Ventolin/Atrovent → O ₂ 99%, increase air entry - Mother not consoled → increasing agitation and becomes obstructive <u>Triggers</u> <i>For progression to next state</i> - Epi, antihistamines, steroids given → 2. Hypotension - 5 minutes → 2. Hypotension
2. Hypotension HR: 140 BP: 60/40 RR: 42 O ₂ SAT: 97% NRB	Patient states he feels dizzy	<u>Learner Actions</u> - ☐ IV NS bolus (20mL/kg) - ☐ Repeat IM epinephrine at 0.01mg/kg/dose - ☐ Start iv epinephrine infusion at 0.1mcg/kg/min	<u>Modifiers</u> - NS bolus → BP 70/50 - Epi infusion → BP 85/60 - No Epi infusion by 5 minutes → nurse suggests it <u>Triggers</u> - Epi infusion started → 3. Airway Obstruction
3. Airway Obstruction O ₂ SAT → 89% over 60 sec. HR: 150 RR: 48 BP: 88/60	Patient complains of throat tightness and tongue swelling	<u>Learner Actions</u> - ☐ Nebulized epinephrine - ☐ Prepare for intubation - ☐ Call Anesthesia/ENT - ☐ Call for difficult airway cart +/- transtracheal jet set-up - ☐ Multiple ETT sizes available - ☐ Consent from mother - ☐ IV Fluid bolus (20mL/kg) - ☐ Push-dose Epi at bedside (give 1mcg/kg bolus q2-3 min) OR increase infusion rate peri-intubation	<u>Modifiers</u> - Epi neb → O ₂ 94% NRB, RR 40 - IV fluid → BP 90/65, HR 130 <u>Triggers</u> - Intubation attempt without paralytic → 4. Laryngospasm - Intubation with paralytic → 6. Resolution - No intubation 4 min into state → 4. Laryngospasm

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4. Laryngospasm HR → to 110 over 1 min O ₂ SAT → 65% over 60 sec. RR → 0 BP → 80 /60	High pitched stridor noise, followed by no visible chest movement.	<u>Learner Actions</u> - ☐ BVM ventilation with PEEP valve (difficult to bag with poor chest rise) - ☐ Update team - ☐ Apply pressure to Larson's point with jaw thrust - ☐ Administer deeper sedation - ☐ Administer paralytic - ☐ Intubate after paralytic	<u>Modifiers</u> - 1 min of BVM → O ₂ SAT to 75% <u>Triggers</u> - Paralytic given → 6. Resolution - 2 min of laryngospasm → 5. Bradycardia
5. Bradycardia O ₂ SAT → 95% over 1 min of bagging HR → 35 over 1 min		<u>Learner Actions</u> - ☐ Give atropine 0.02mg/kg iv (0.56mg) - ☐ Continue bagging patient if not yet intubated	<u>Modifiers</u> <u>Triggers</u> - Atropine given → 6. Resolution - 13 minutes → 6. Resolution
6. Resolution HR: 100/min BP: 90/50 RR: 10 (vented) O ₂ SAT: 100%		<u>Learner Actions</u> - ☐ Post-intubation sedation - ☐ ICU Consult - ☐ Update mother	<u>Modifier</u> - Mother not updated → gets very upset. Questions care. Threatens to sue. <u>Triggers</u> - ICU consult → END CASE

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Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results

No blood work required.

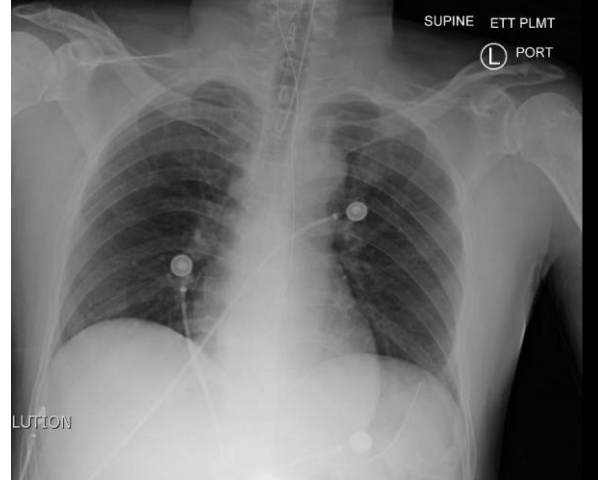
Images (ECGs, CXRs, etc.)

Initial CXR - normal pediatric CXR



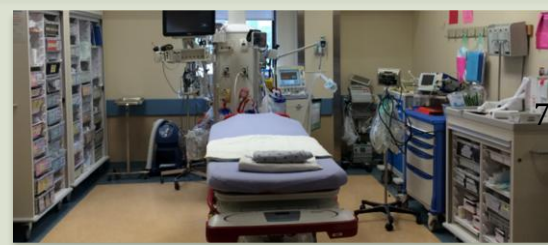
CXR source: <http://radiology-information.blogspot.ca/2015/04/normal-chest-x-ray.html>

Post-intubation - normal



CXR source: <http://jetem.org/ettcxr/>

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
Educational Goal:	To expose learners severe anaphylaxis and laryngospasm complication during intubation.
CRM Objectives:	1) Displays leadership by maintaining calm demeanor during crisis and acting decisively 2) Employs good communication skills by using closed loop, listening to the input of others, and addressing concerned family members
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Sample Questions for Debriefing	
1) When did you decide to switch over to IV Epinephrine? What are its indications? 2) What was your approach to pending airway obstruction? What was your back-up method of securing the airway? 3) What is your approach to laryngospasm? 4) What is Larson's point? 5) What are some other complications of sedation with ketamine? 6) How did it feel as a team to manage this extremely unexpected complication? Do you feel you remained calm? 7) How did the leader address the critical nature of this complication? As a team, did you feel that you had clear directions?	
Key Moments	
Recognition and treatment of anaphylaxis and it's complications including bronchospasm, hypotension and airway obstruction	
Recognition of Laryngospasm	
Recognition of hypoxia-mediated bradycardia	